M365

Fresh Tenant Setup

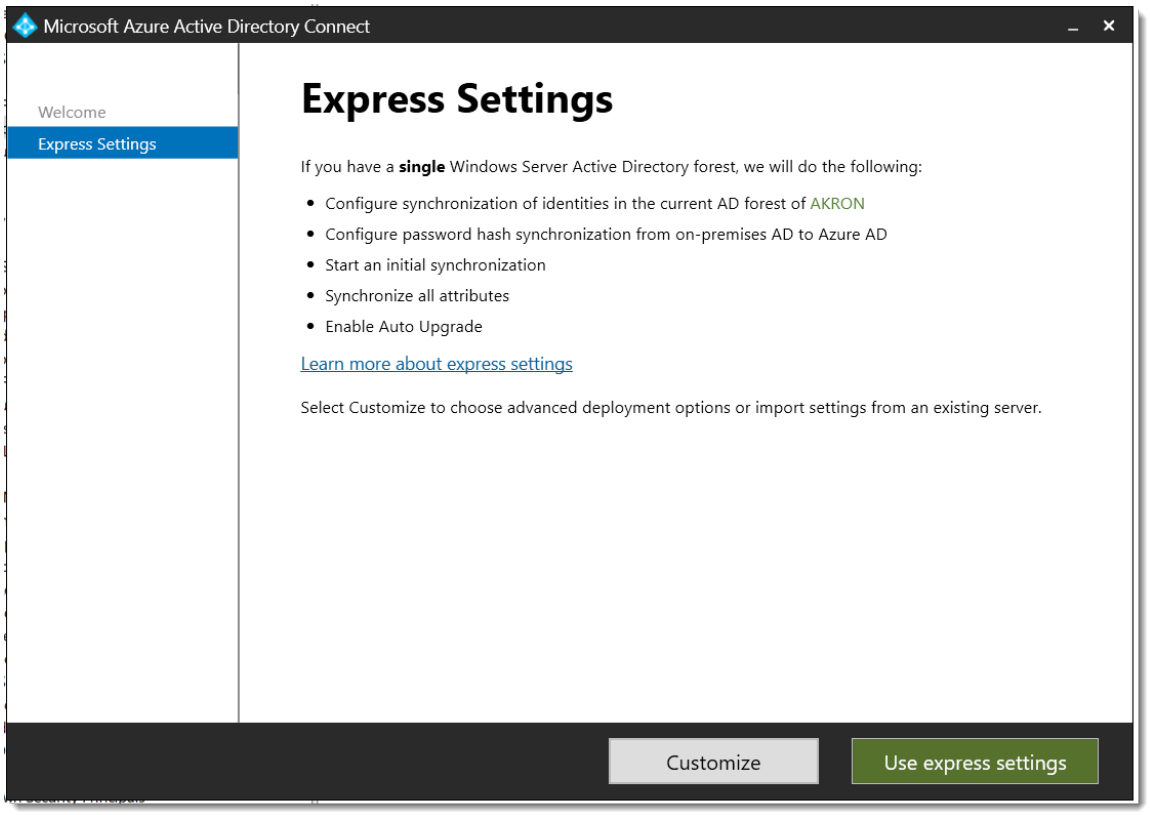
These are the steps I typically take to set up a fresh M365 E5 tenant. This is not complete. I'm documenting this (finally) as I rebuild a new test tenant.

Entra ID

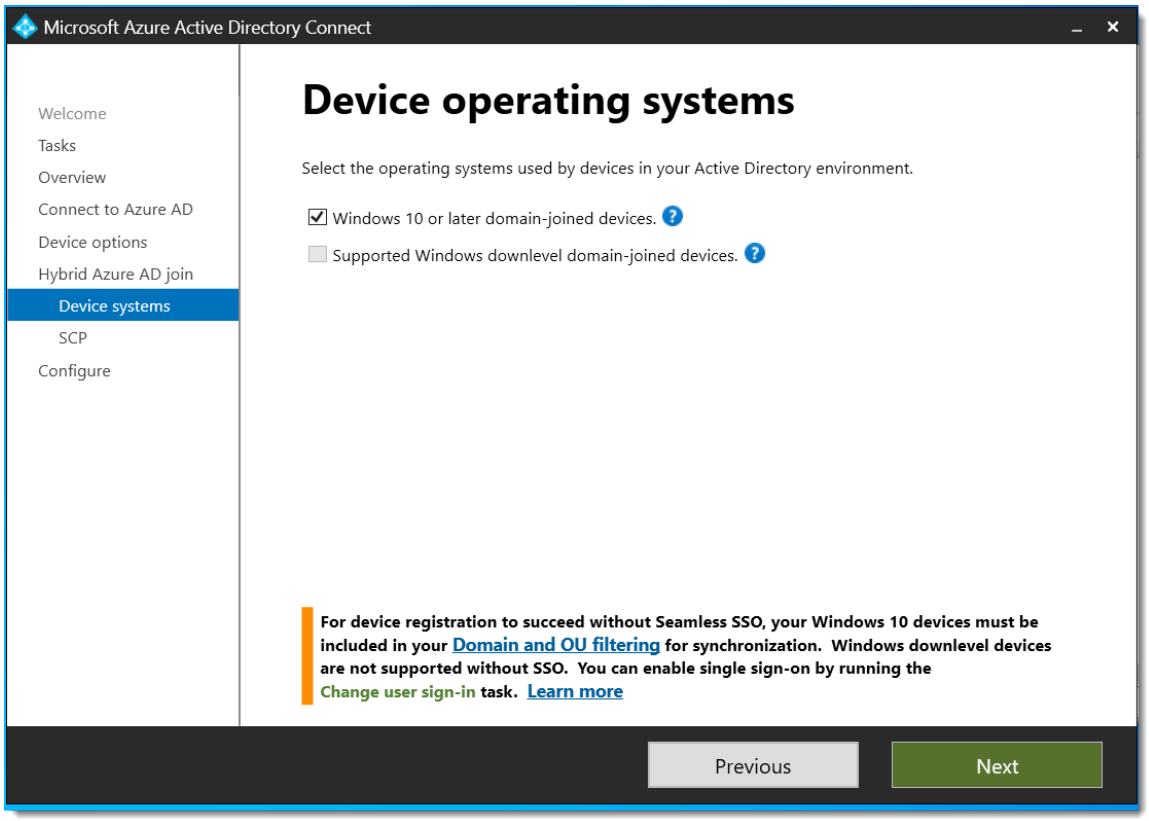
Azure AD Connect

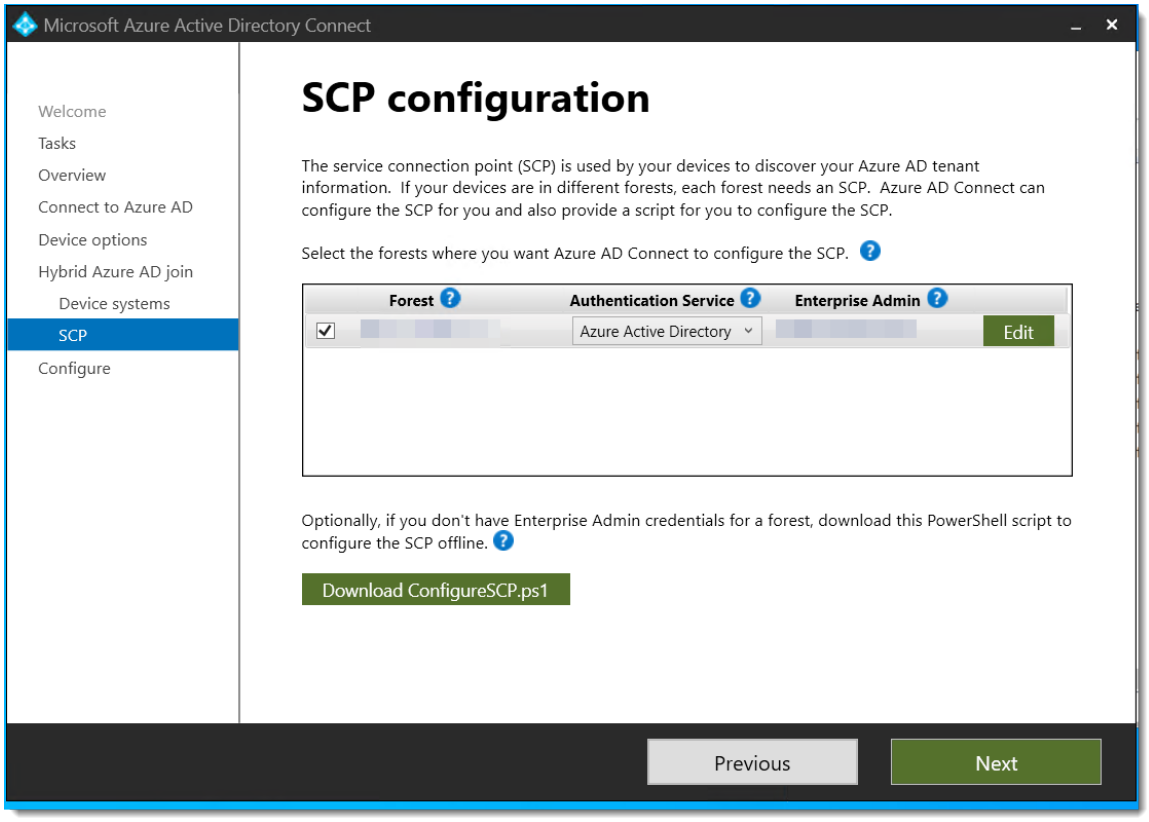
Assuming you already have a Lab Domain Controller and Server for AAD Connect.

Install AAD Connect, use Express Settings



After it is finished, re-run and enable Hybrid Join





Enable Group Writeback

Docs

On the AzureAD Connect server, run the following commands to enable Group Writeback in Windows PowerShell

```
Set-ADSyncScheduler -SyncCycleEnabled $false
Set-ADSyncAADCompanyFeature -GroupWritebackV2 $true
Set-ADSyncScheduler -SyncCycleEnabled $true
Start-ADSyncSyncCycle -PolicyType Initial
```

Hybrid Cloud Trust

📘 Docs

Create EntraID Kerberos Server

```
# Specify the on-premises Active Directory domain. A new Azure AD
# Kerberos Server object will be created in this Active Directory domain.
$domain = "contoso.com"

# Enter a UPN of an Azure Active Directory global administrator
$userPrincipalName = "admin@contoso.com"

# Enter a domain administrator username and password.
$domainCred = Get-Credential

# Create the new Azure AD Kerberos Server object in Active Directory
# and then publish it to Azure Active Directory.
# Open an interactive sign-in prompt with given username to access the Azure
AD.
Set-AzureADKerberosServer -Domain $domain -UserPrincipalName
$userPrincipalName -DomainCredential $domainCred

# Verify server
Get-AzureADKerberosServer -Domain $domain -DomainCredential $domainCred -
UserPrincipalName $userPrincipalName

Id                  : 17530
UserAccount         : CN=krbtgt_AzureAD,CN=Users,DC=contoso,DC=com
ComputerAccount     : CN=AzureADKerberos,OU=Domain
Controllers,DC=contoso,DC=com
DisplayName         : krbtgt_17530
DomainDnsName       : contoso.com
KeyVersion          : 27591
KeyUpdatedOn        : 10/13/2022 9:23:43 PM
KeyUpdatedFrom      : CONTOSO-DC-01.contoso.com
CloudDisplayName    : krbtgt_17530
CloudDomainDnsName  : contoso.com
CloudId             : 17530
CloudKeyVersion     : 27591
CloudKeyUpdatedOn   : 10/13/2022 9:23:43 PM
CloudTrustDisplay   :
```

Entra

Identity Protection

Self Service Password Reset

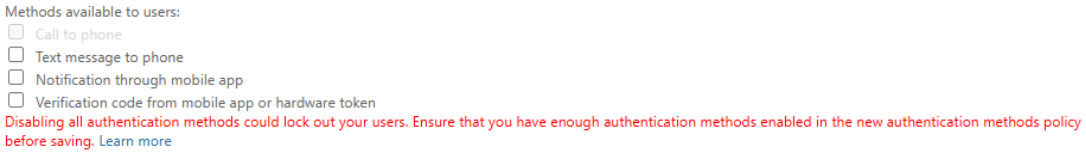
[🔗 Portal]
📘 Docs

- Enable Self service password reset
  - Target a group that excludes service accounts.

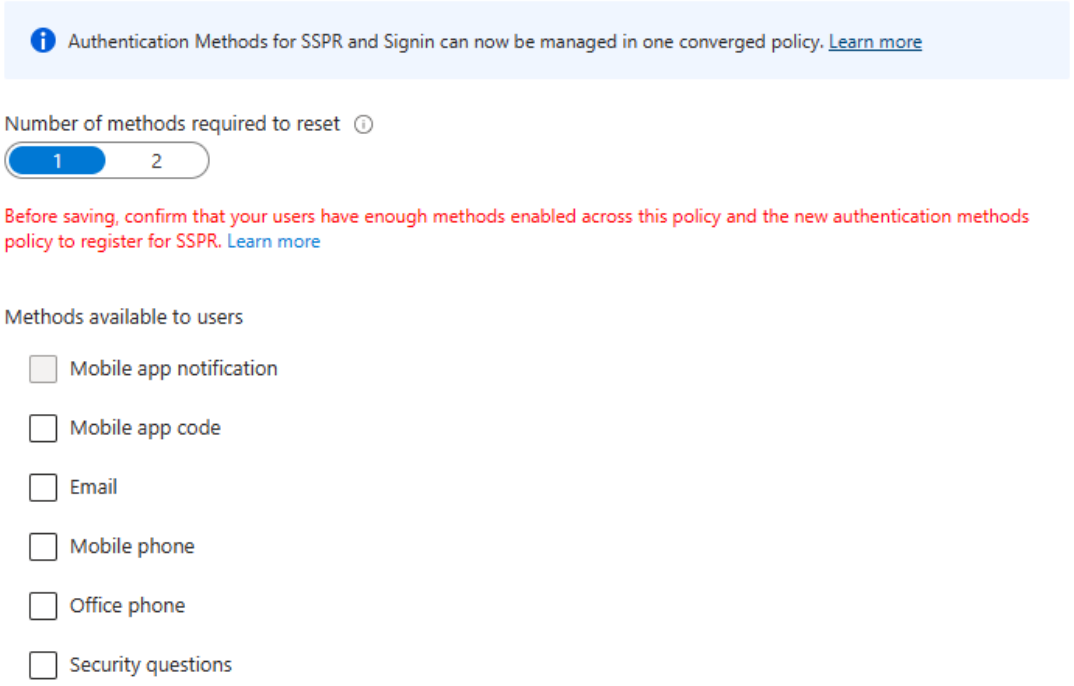
Authentication Methods

🔗 Portal
📘 Docs

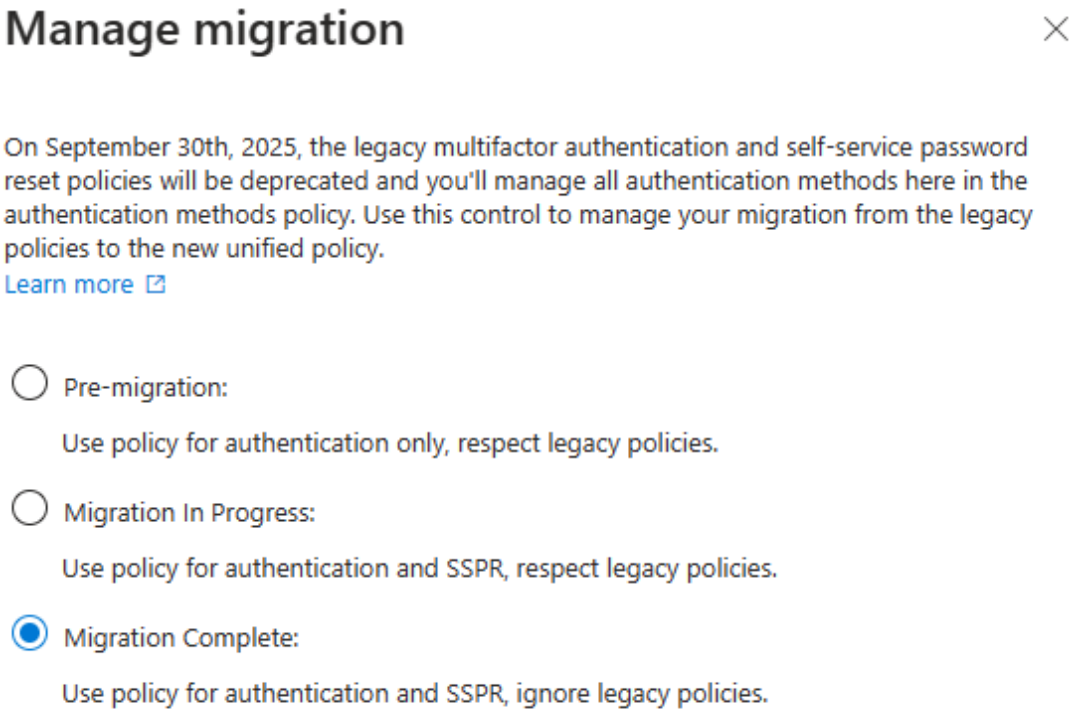
- Enable FIDO2, Authenticator, Temporary Access Pass in Authentication Methods
- Disable verification options in the legacy MFA settings portal



- Disable Authentication methods in SSPR Authentication Methods



- Migrate to the Converged Authentication Methods Policy

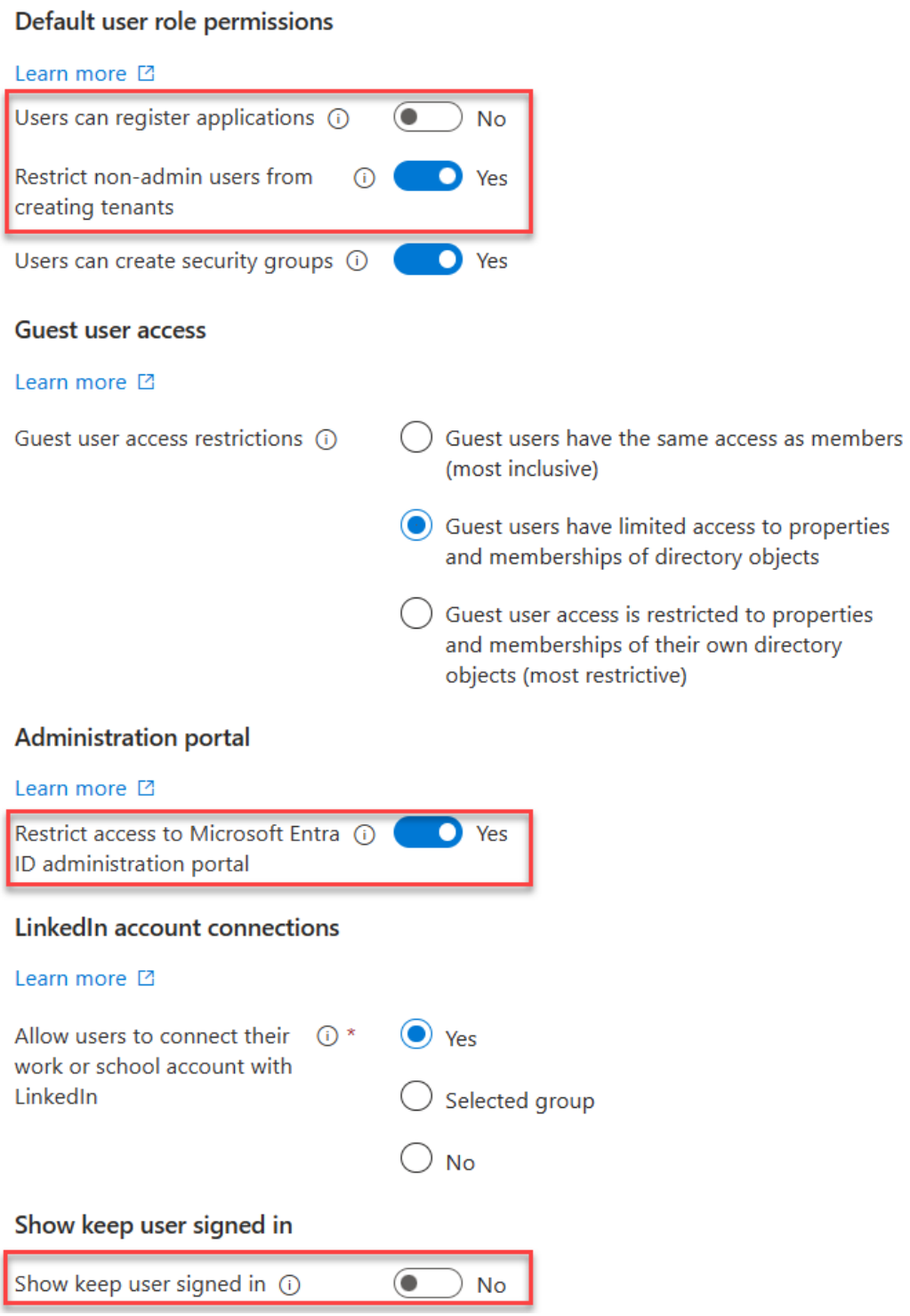


User Settings

🔗 Portal

- Toggle Off
  - Users can register Applications
  - Show keep user signed in
- Toggle On
  - Restrict non-admin users from creating tenants

- Restrict access to Entra ID administration portal



User Feature Settings

🔗 Portal

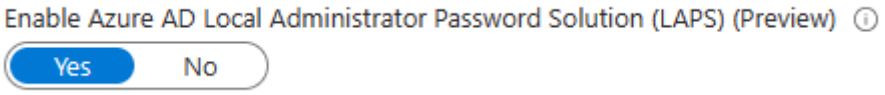
- Select All for Users can use preview features for My Apps

Device Settings

Cloud LAPS

Portal
Docs

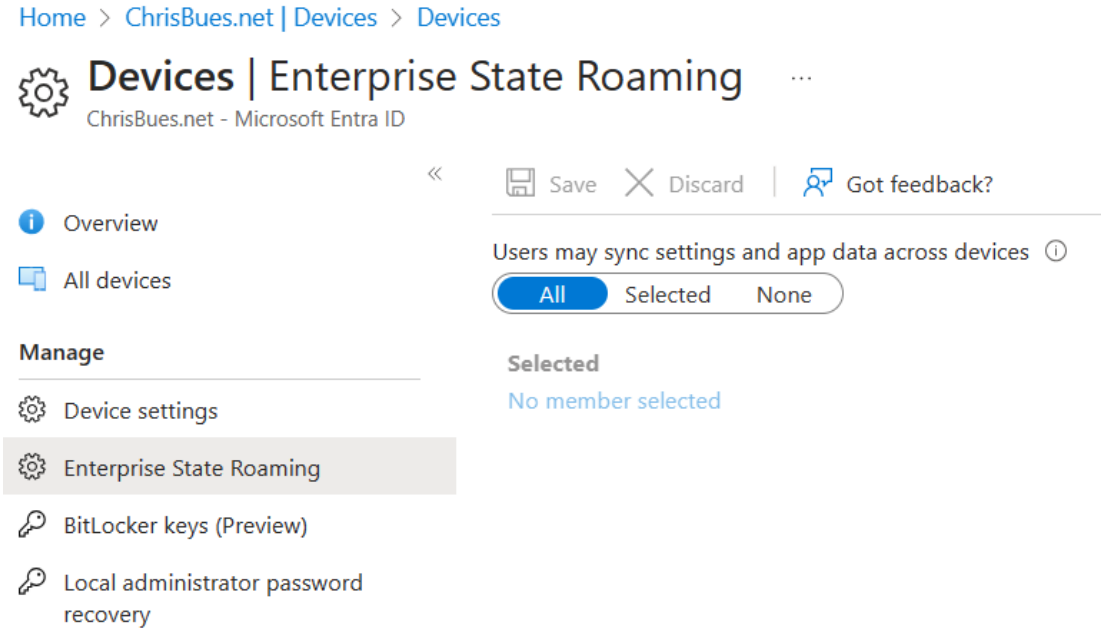
- Enable LAPS



Enterprise State Roaming

Portal

- Enable Enterprise State Roaming



Identity Protection

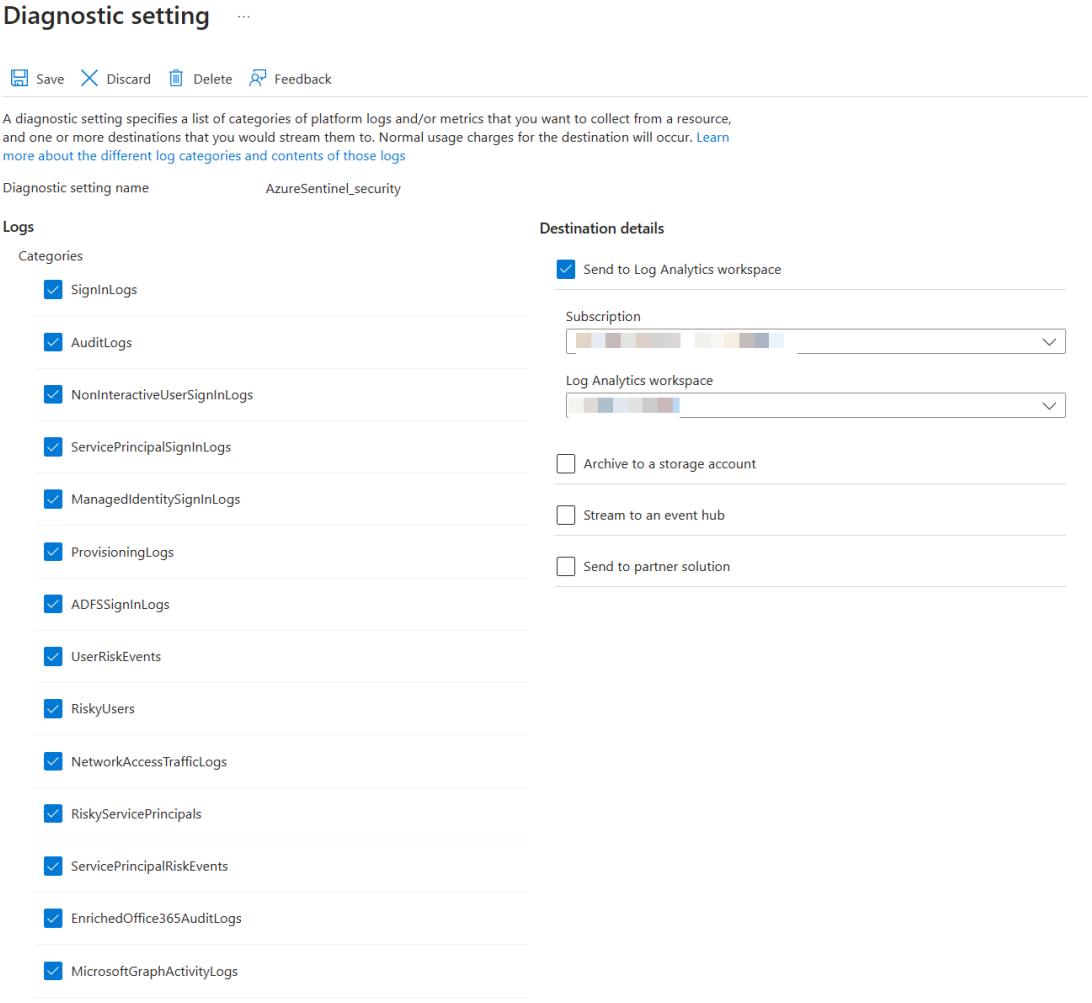
Multifactor authentication registration policy

- Create a AAD group called Service Accounts, add the AzureAD Connect sync account
- Enable the policy, targeting all users and excluding

Diagnostic Settings

Portal
Docs

- Enable all diagnostic settings to log to your Sentinel's log analytics workspace

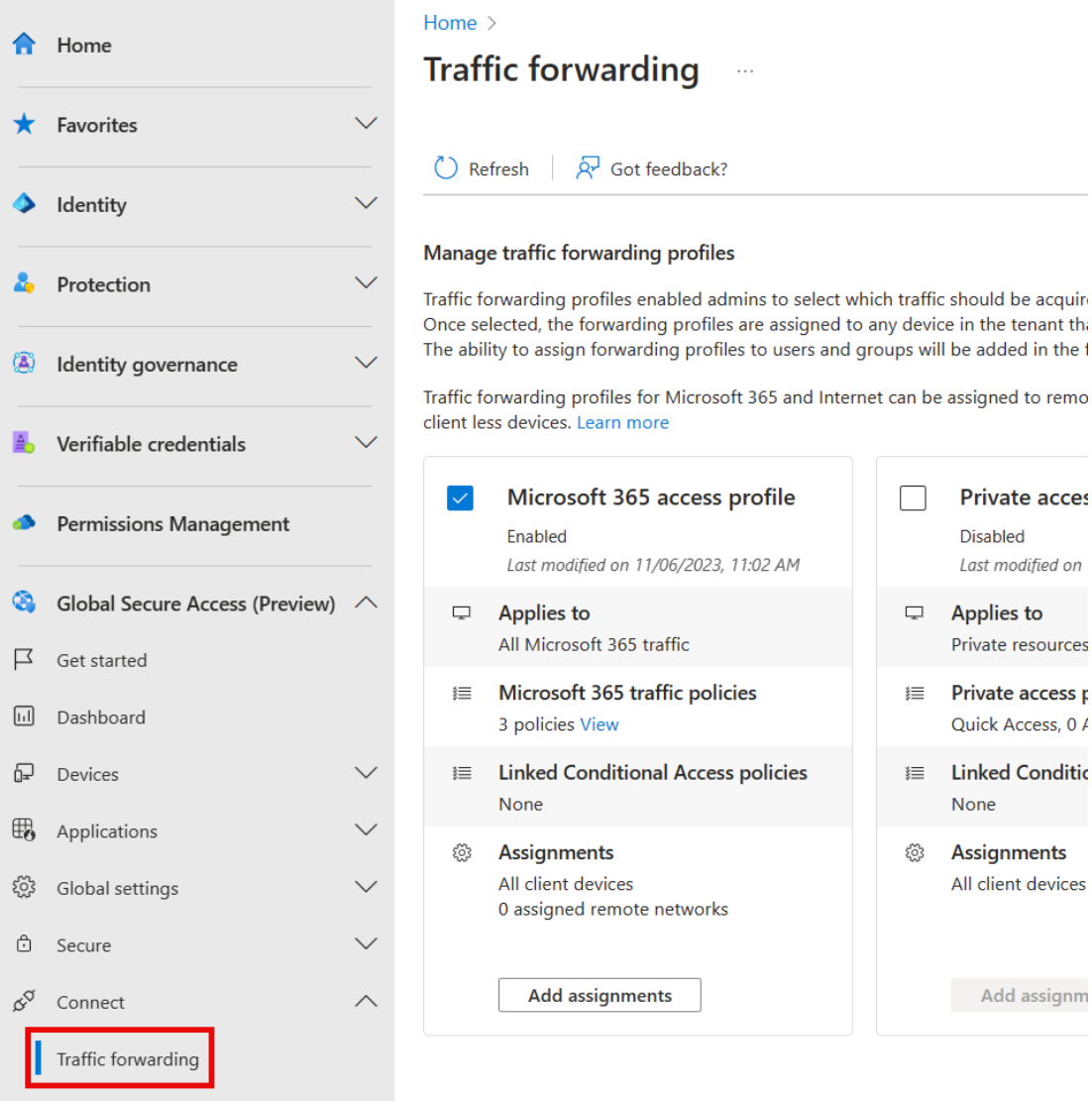


Global Secure Access

🔗 Portal
📘 Docs

Internet Access

1. Enable the M365 Profile



2. Download the GSA Client and deploy to Windows devices.

Intune

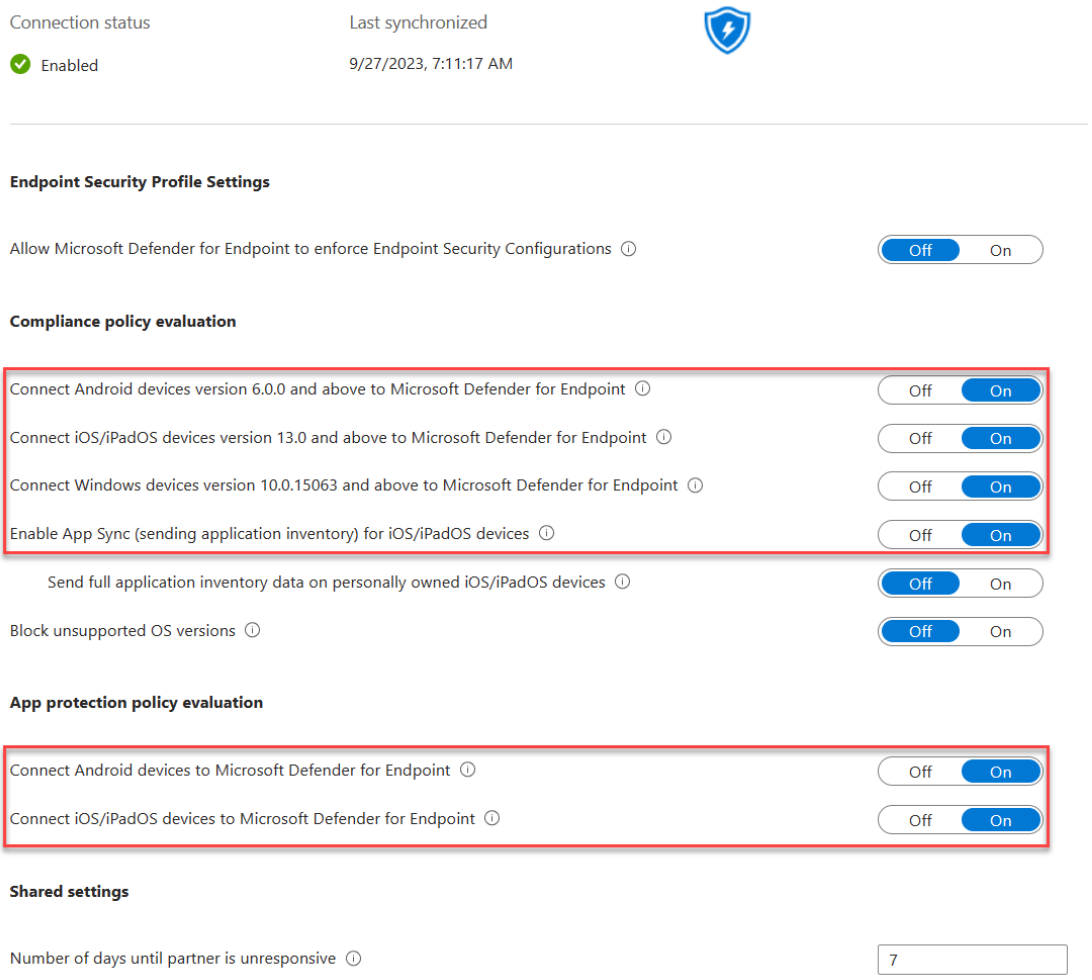
Tenant Administration Settings

Defender for Endpoint Connector

🔗 Portal

- Enable compliance policy evaluation for all platforms
- Enable app sync

- Enable App protection policy evaluation

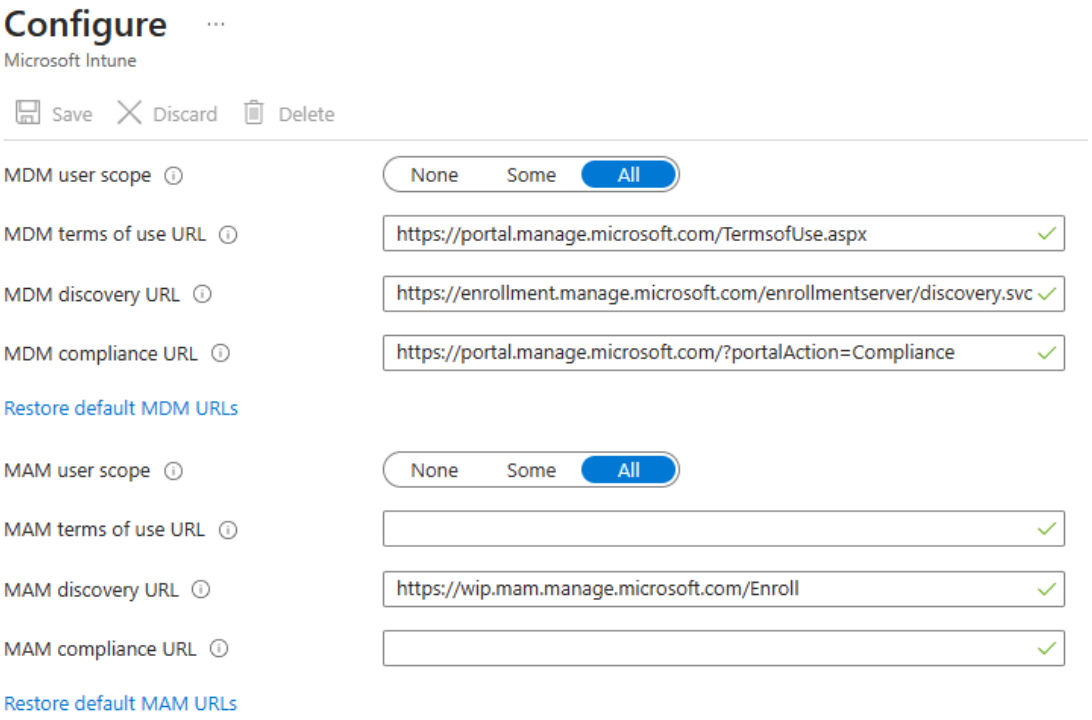


Devices

Windows Automatic Enrollment

🔗 Portal

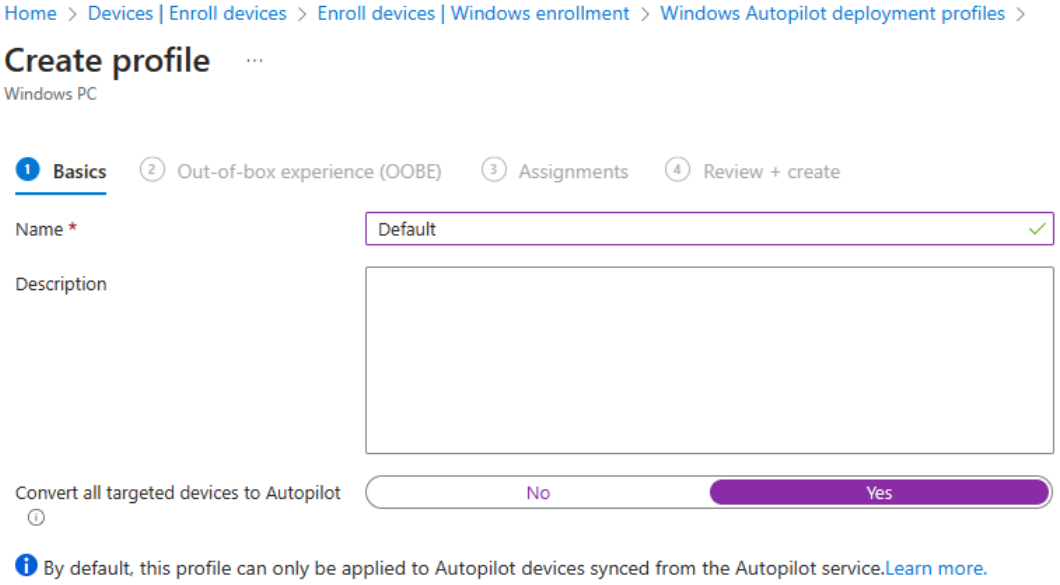
- Set MDM and MAM user scopes to all



Windows Autopilot

Portal

- Create a new Autopilot deployment profile



Basics | 2 Out-of-box experience (OOBE) | 3 Assignments | 4 Review + create

Configure the out-of-box experience for your Autopilot devices

Deployment mode *	User-Driven
Join to Azure AD as *	Azure AD joined
Microsoft Software License Terms	Show / **Hide**

Important information about hiding license terms

Privacy settings	Show / **Hide**

The default value for diagnostic data collection has changed for devices running Windows 10, version 1903 and later, or Windows 11.

Hide change account options	Show / **Hide**
User account type	**Administrator** / Standard
Allow pre-provisioned deployment	**No** / Yes
Language (Region)	Operating system default
Automatically configure keyboard	No / **Yes**
Apply device name template	No / **Yes**

Create a unique name for your devices. Names must be 15 characters or less, and can contain letters (a-z, A-Z), numbers (0-9), and hyphens. Names must not contain only numbers. Names cannot include a blank space. Use the %SERIAL% macro to add a hardware-specific serial number. Alternatively, use the %RAND:x% macro to add a random string of numbers, where x equals the number of digits to add.

Enter a name * -%RAND:4%

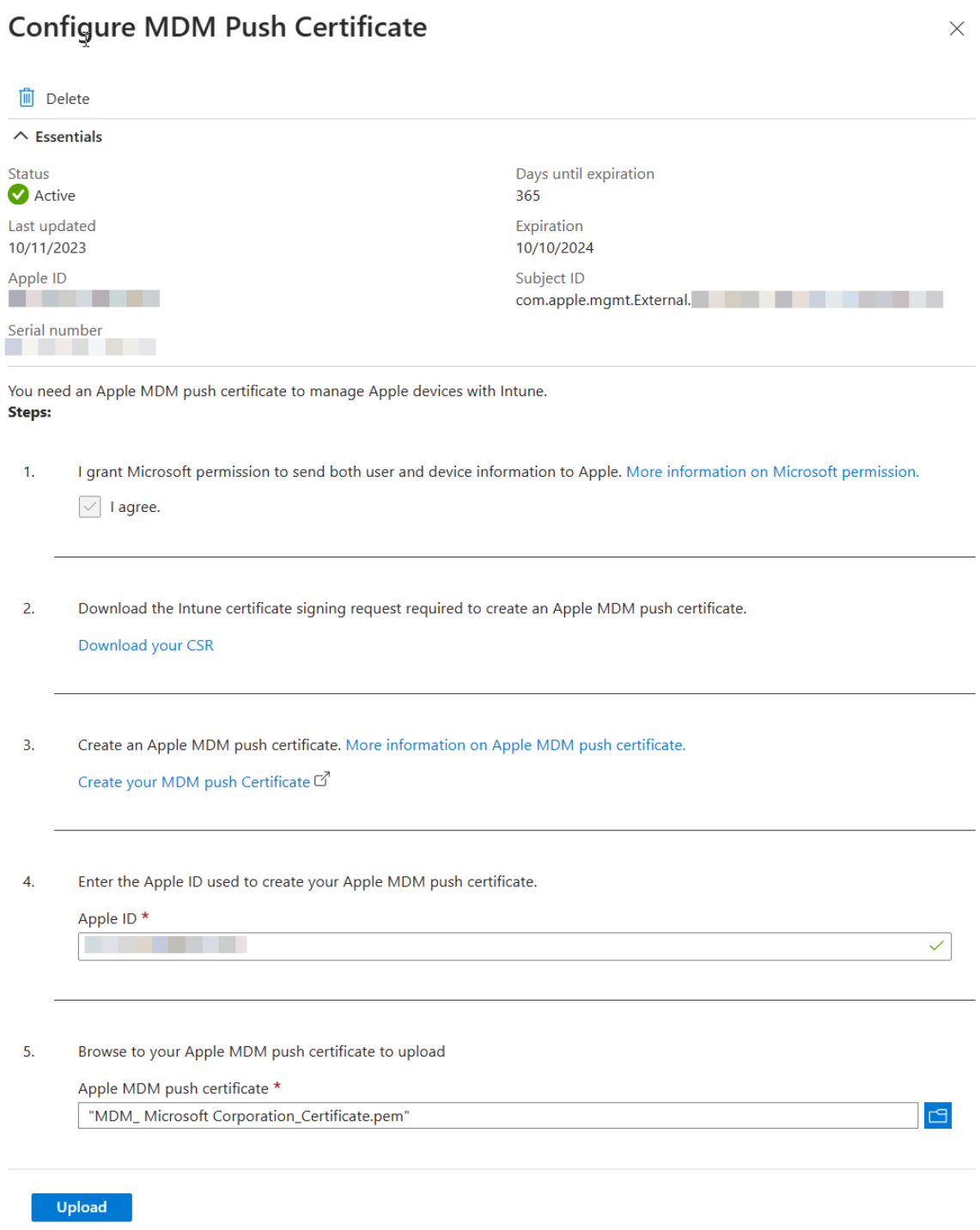
iOS Enrollment

Portal

User-driven iOS enrollment is a two step process - the push certificate and the enrollment profile.

CONFIGURE APPLE MDM PUSH CERTIFICATE

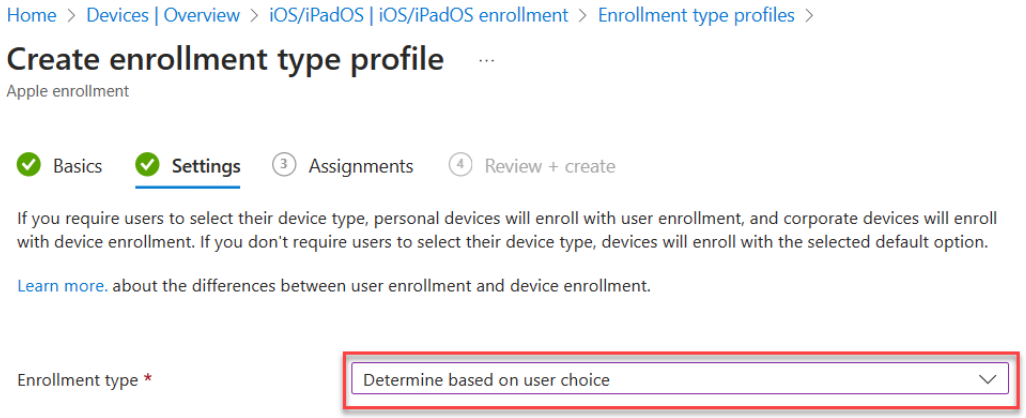
Docs



ENROLLMENT PROFILE

Portal

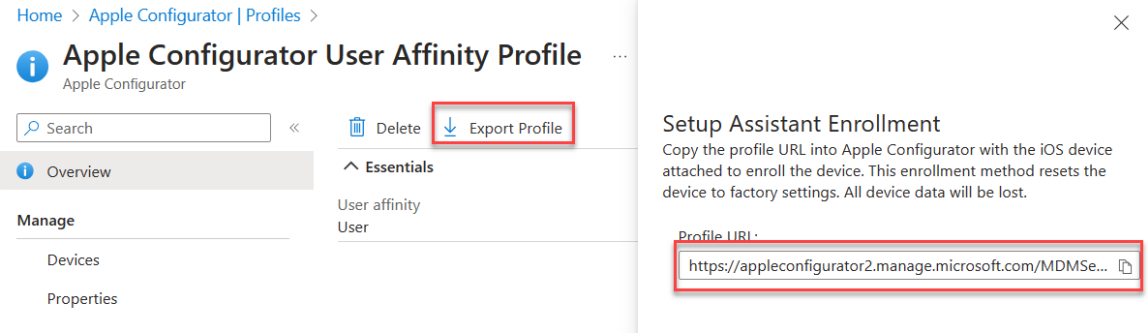
- Configure an enrollment profile
  - Create a profile that allows user choice of type of device (corporate vs user), target all users.



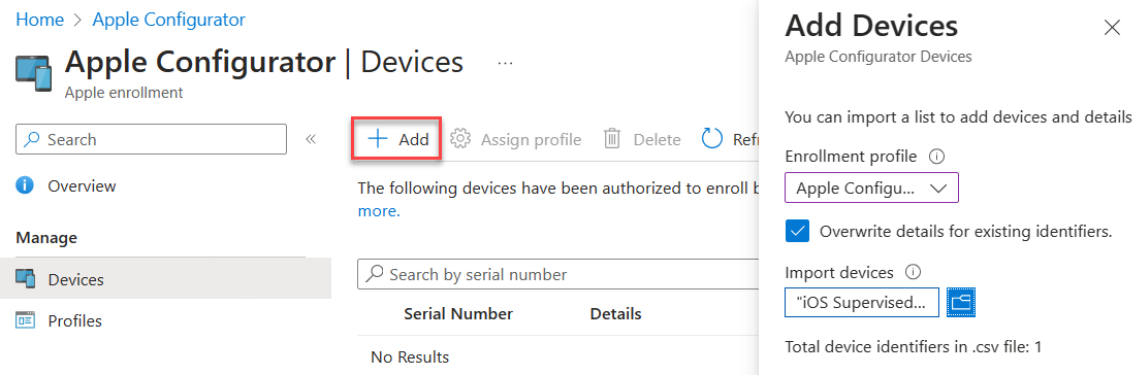
SUPERVISED IOS ENROLLMENT WITH APPLE CONFIGURATOR

Portal

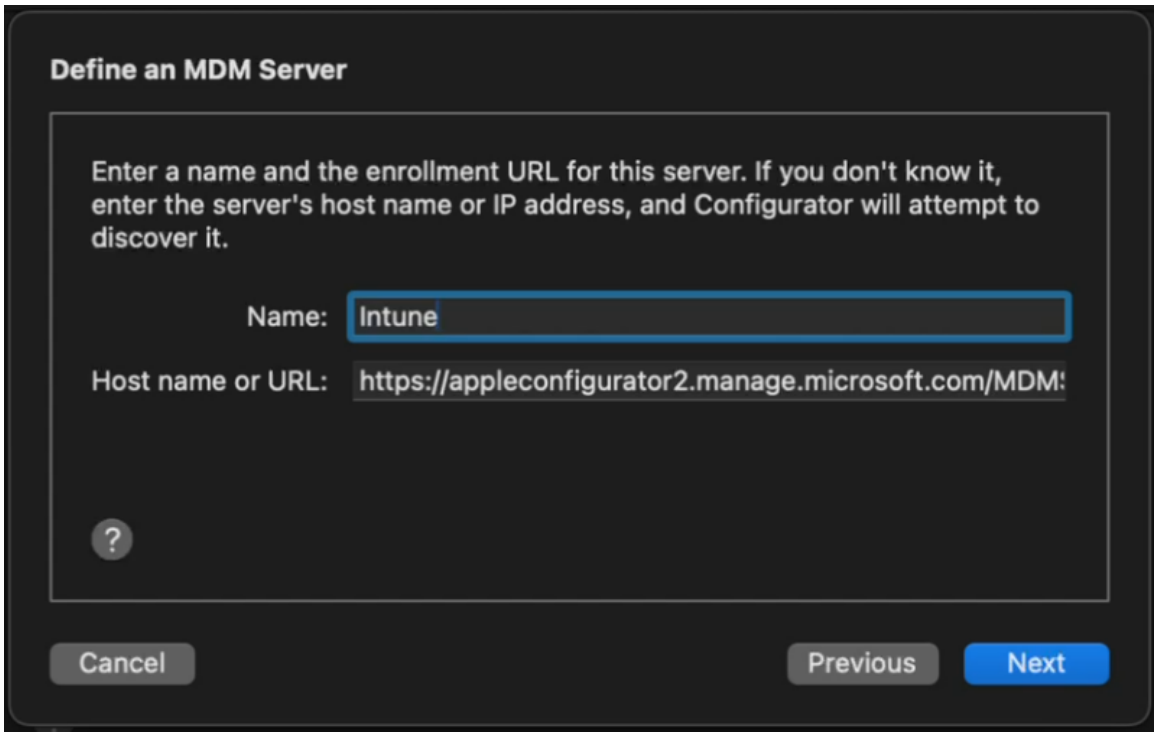
Docs There are two options for Apple Config profile - with user affinity and without. For testing, enrollment with User Affinity with the Company Portal app mimics how devices might be distributed to end users. 1. Create a new Enrollment Profile. On the settings step, select: User affinity: `Enroll with User Affinity` Select where users must authenticate: `Company Portal` 2. Export the profile you just created. Copy the URL.



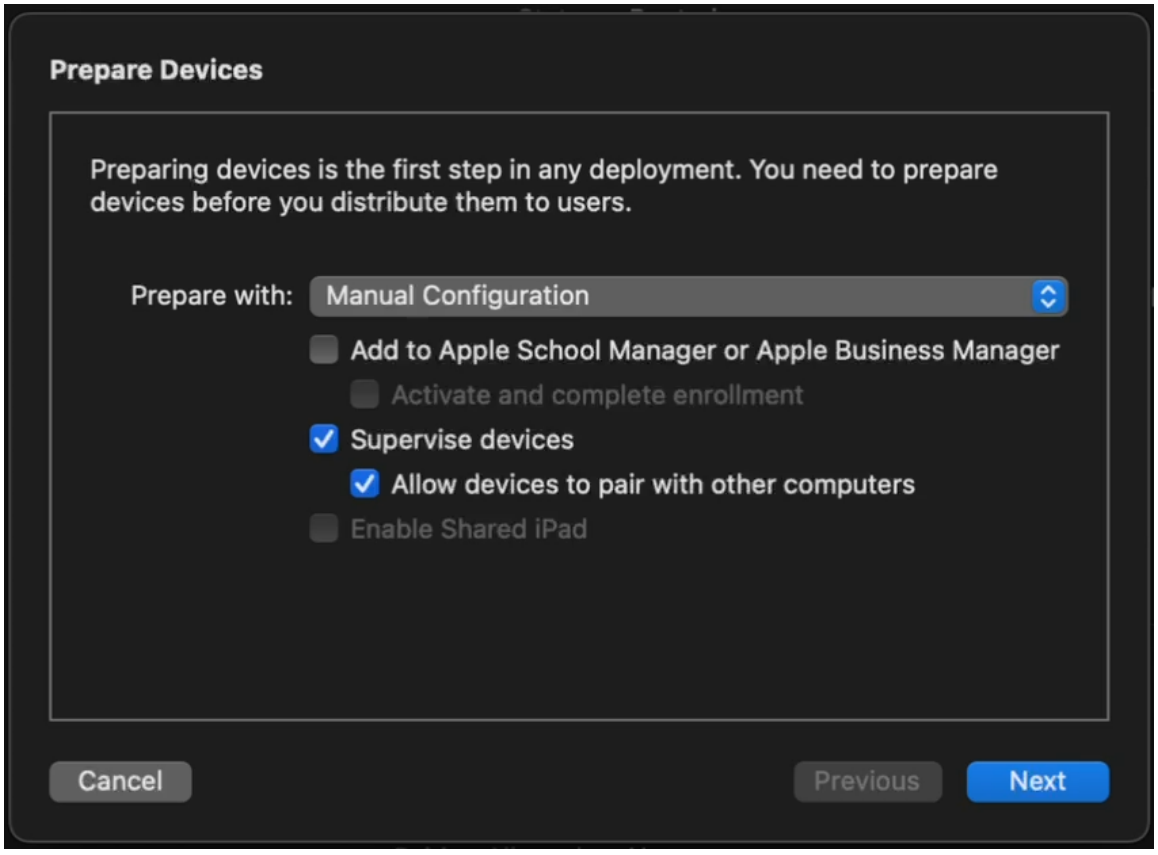
3. Create a csv file with the serial numbers of iPads you wish to enroll. `Serial number,device details` 4. Upload the csv file in the portal under Devices. Assign the profile you just created.



5. In Apple Configurator, choose Settings -> Servers. Click + to add a server. Add the URL you copied from step 2.



6. Connect a device, and at the main screen, click Prepare. Leave the default options unchanged.



7. Choose the Intune MDM server defined in Step 5.

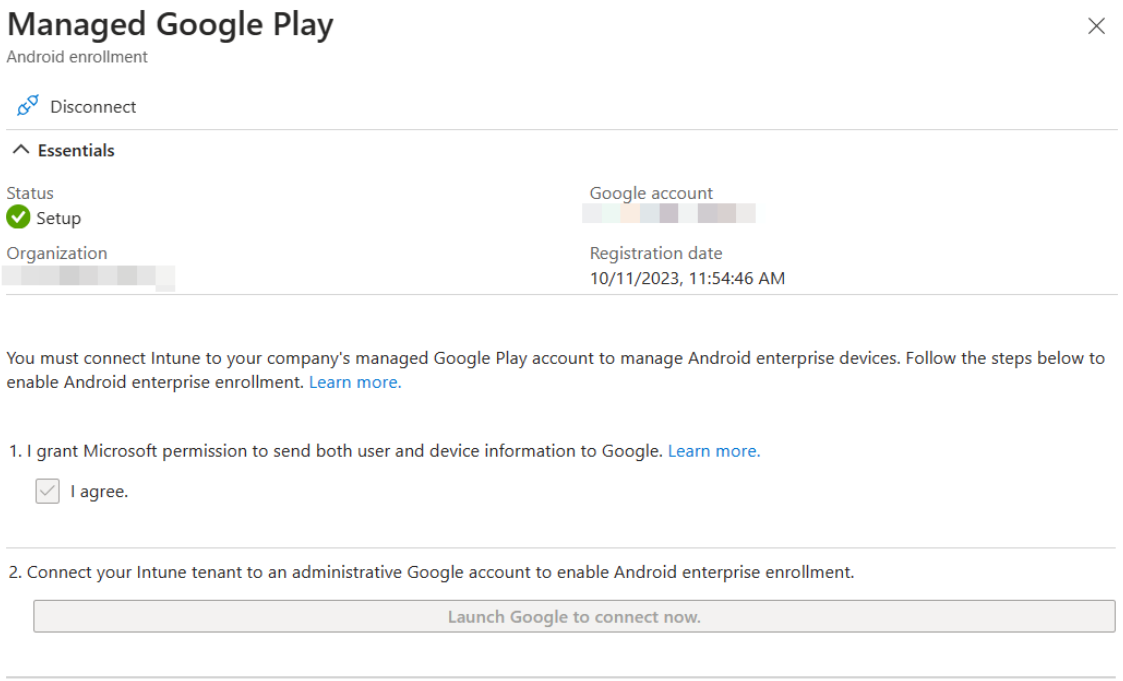
8. Skip Apple Business Manager sign-in if prompted. At the Organization screen select a previous org or create a new one. This is shown in the settings app in iOS. 9. Choose to generate a new supervision Identity or reuse an existing one. 10. Choose which steps to display in the Setup Assistant. Click Prepare to start the process.

Android Enrollment

🔗Portal
📘Docs

User-driven Android enrollment is a two step process - the managed Google Play account linking and the enrollment profile.

MANAGED GOOGLE PLAY ACCOUNT LINKING

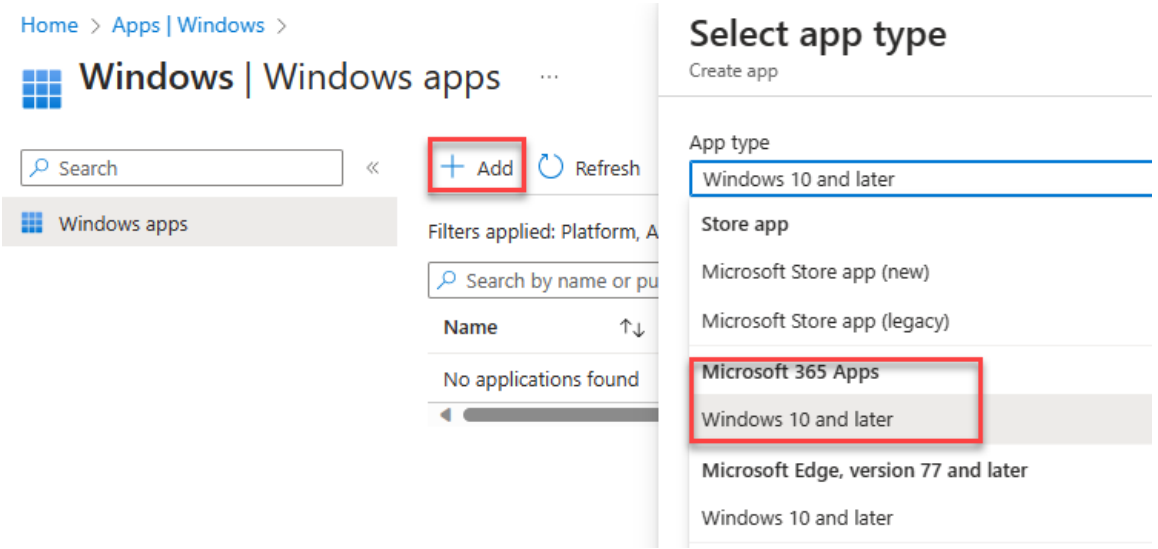


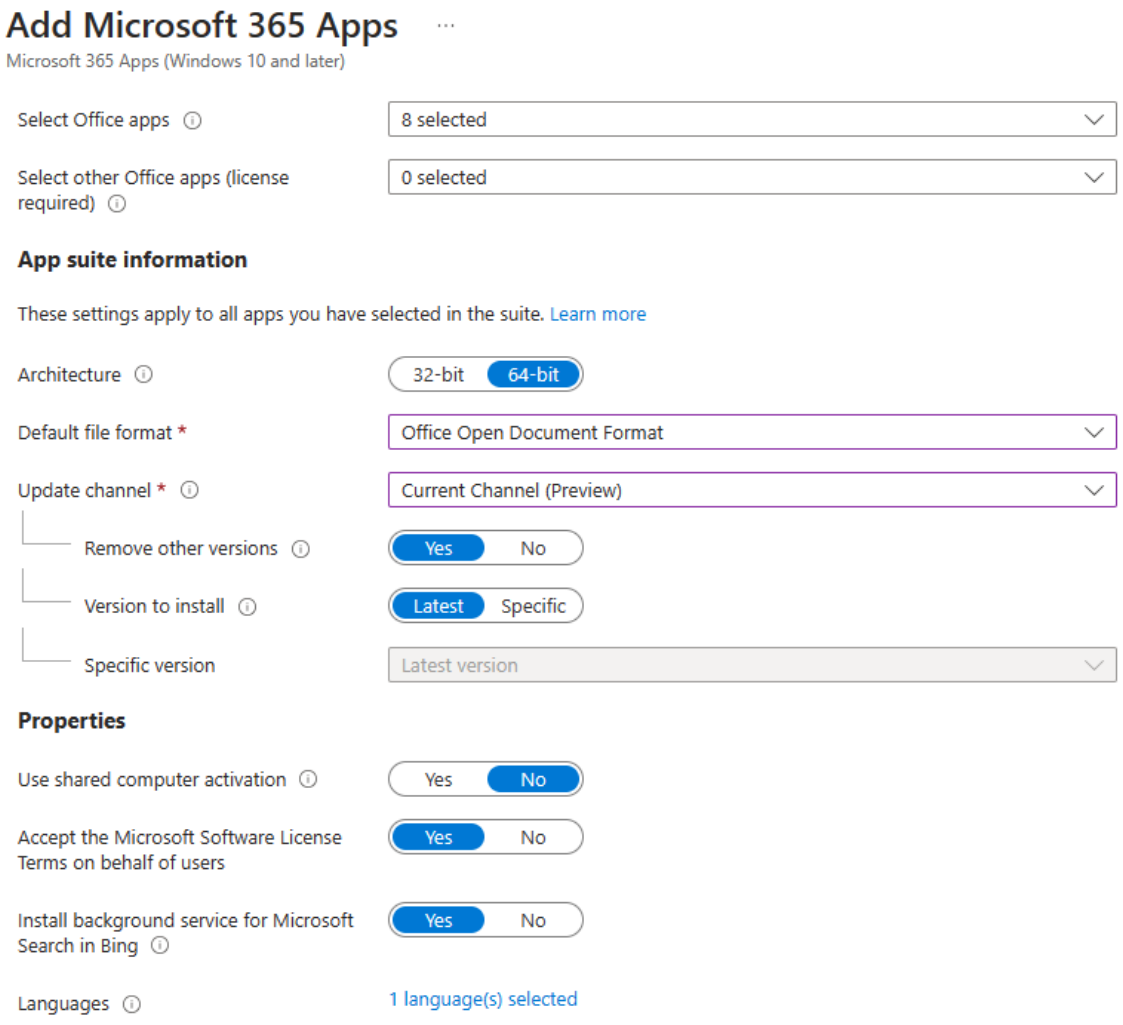
Applications

🔗Portal
📘 Docs

Windows

Add app -> Microsoft 365 Apps for Windows 10 and Later. Assign to all devices.



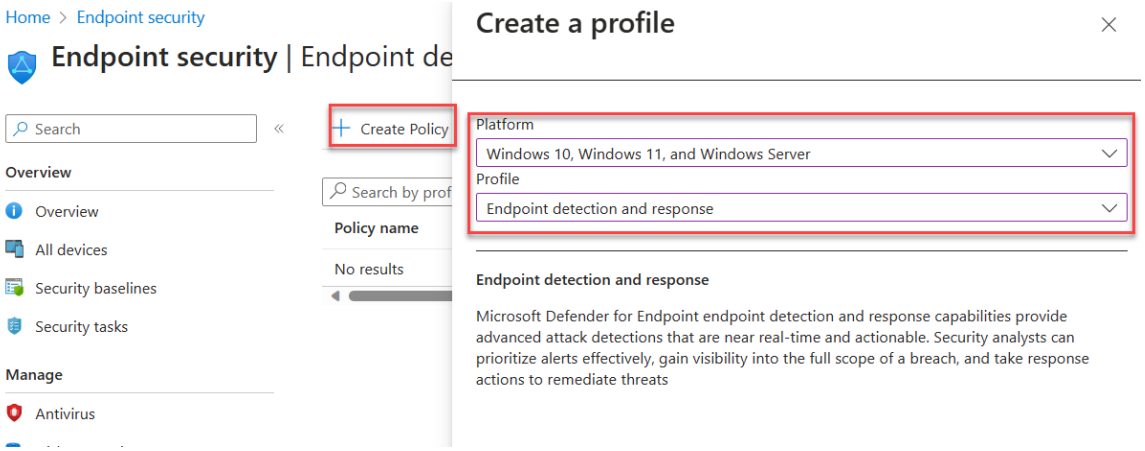


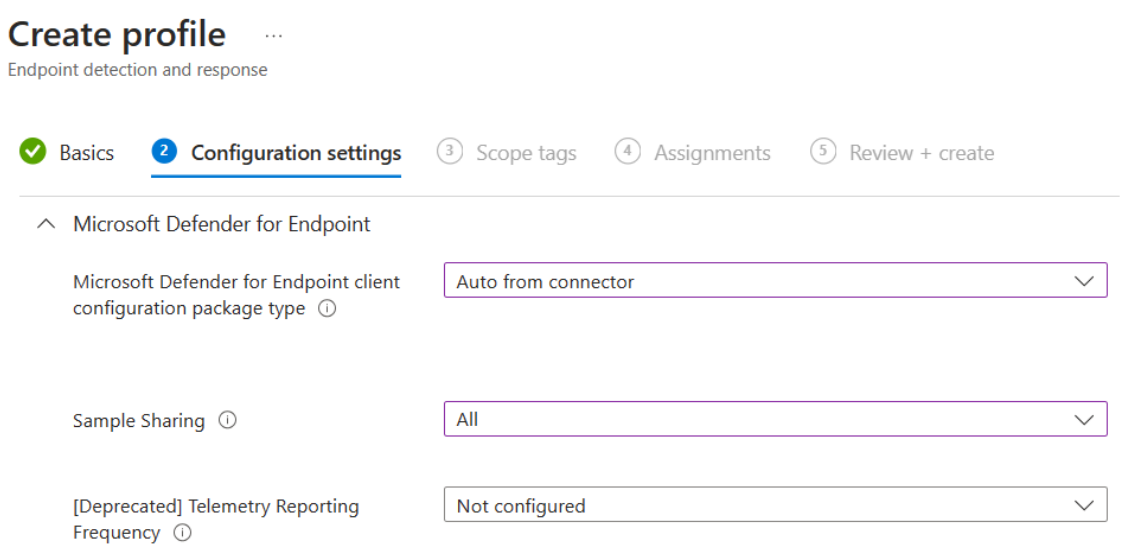
Endpoint security

🔗Portal

WINDOWS

Under Endpoint Detection and Response, create a new policy targeting Windows 10,11 and Server. Target all devices.





MacOS

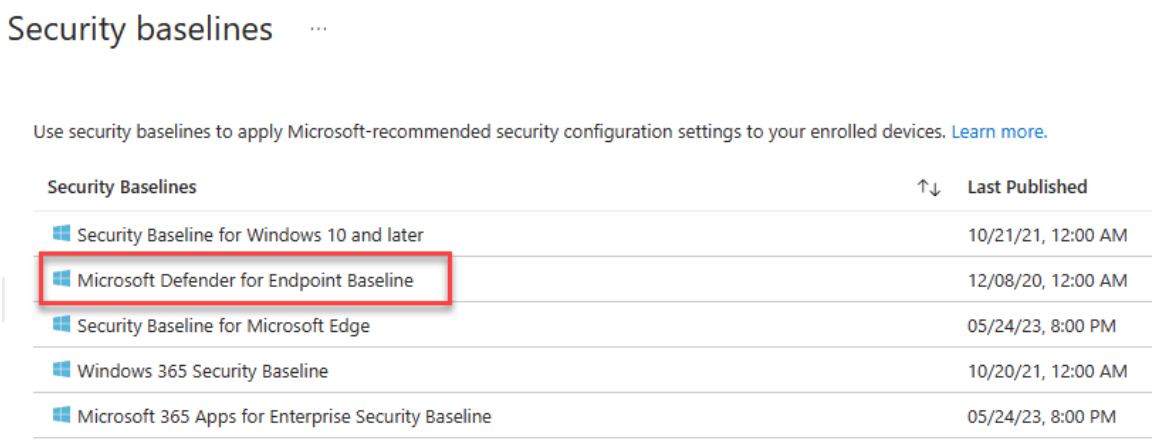
Deploying MDE on MacOS is a multi-step manual process, and changes occasionally. Refer to Intune-based deployment for Microsoft Defender for Endpoint on Mac - Microsoft Defender for Endpoint | Microsoft Learn for the most current steps.

The config files for MDE on Mac and Linux are in this GitHub repo.

1. Approve system extensions
2. Deploy the combined mobileconfig file from the repo to simplify policies.
3. Deploy the Microsoft AutoUpdate config file from the repo
4. Deploy the MDE Preferences config file from the repo to enable Endpoint DLP.
5. Deploy the Defender App via Intune
6. Deploy the MDE onboarding package

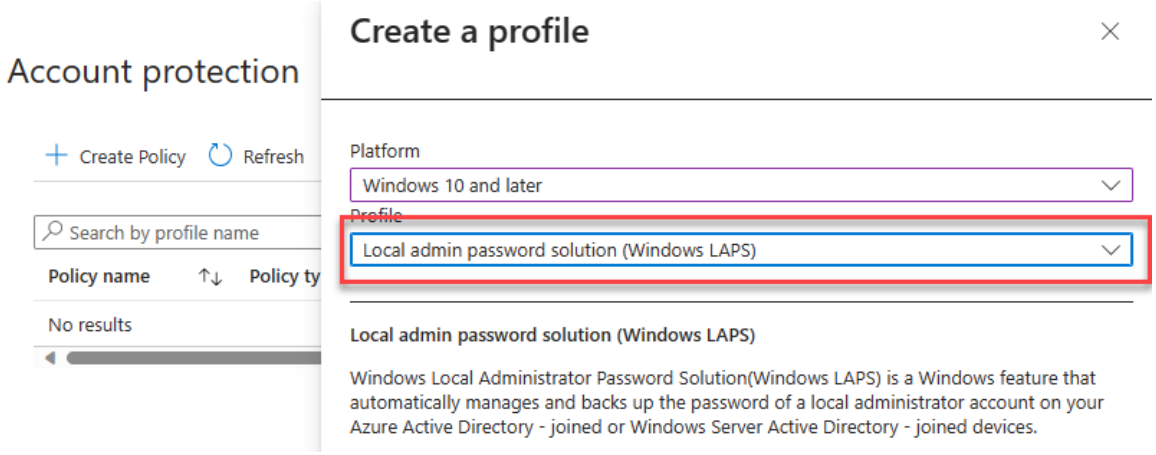
Security Baselines

Create a new Microsoft Defender for Endpoint Baseline policy and target all devices.



Account Protection

After enabling LAPS in Entra, create a Windows LAPS profile and apply to all devices.



Home > Endpoint security | Account protection >

Create profile

Local admin password solution (Windows LAPS)

Basics | 2 Configuration settings | 3 Scope tags | 4 Assignments | 5 Review + create

LAPS

Backup Directory	Backup the password to Azure AD only
Password Age Days	Not configured 30
Administrator Account Name	Not configured
Password Complexity	Not configured
Password Length	Not configured
Post Authentication Actions	Not configured
Post Authentication Reset Delay	Not configured

M365 Defender

Email & collaboration

Policies & Rules -> Threat Policies -> Preset Security Configurations

Enable Standard Protection Preset Policies.

Standard protection

A baseline protection profile that protects against spam, phishing, and malware threats.

- ✓ Balanced actions for malicious content
- ✓ Balanced handling of bulk content
- ✓ Attachment and link protection with Safe Links and Safe Attachments

Standard protection is off

Manage protection settings

Apply Exchange Online Protection

Add the users, groups, and domains to protect using Exchange Online Protection capabilities, including inbound anti-spam, anti-malware, and anti-phishing. Learn more about preset security policies

Apply protection to:

- (●) All recipients
- ( ) Specific recipients
- ( ) None

Apply Defender for Office 365 protection

Add the users, groups, and domains to protect using Defender for Office 365 capabilities, including Safe Attachments and Safe Links. Learn more about preset security policies

Apply protection to:

- ( ) Previously selected recipients
- (●) All recipients
- ( ) Specific recipients
- ( ) None

MDCA

In the Defender Portal, go to Settings -> Cloud Apps

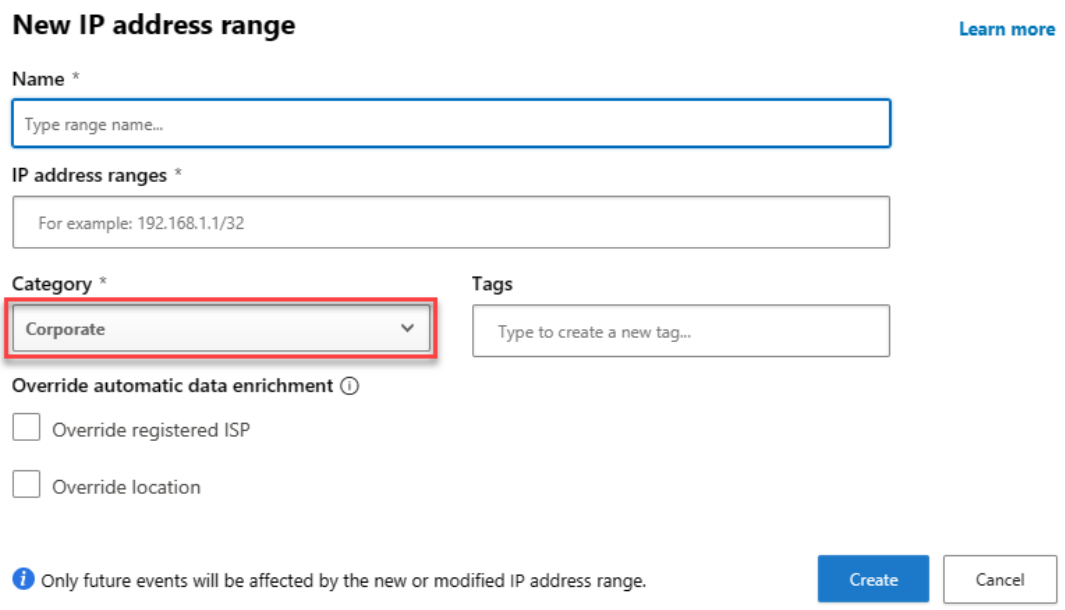
System

PREVIEW FEATURES

- Toggle Enable

IP ADDRESS RANGES

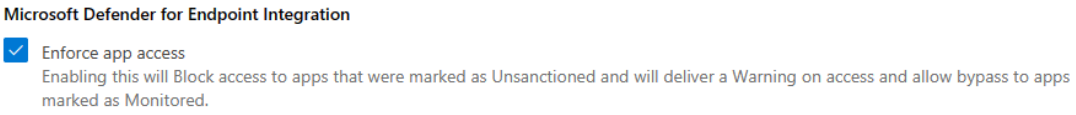
- If you have IP Ranges as Trusted Named Locations in EID, add them as Custom IP Address Ranges in MDCA with the category of Corporate



Cloud Discovery

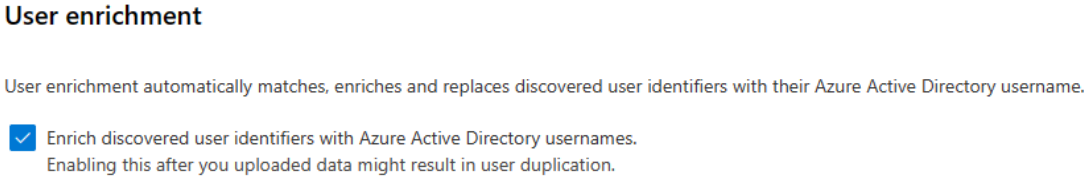
DEFENDER FOR ENDPOINT

- Enforce App Access with Defender for Endpoint



USER ENRICHMENT

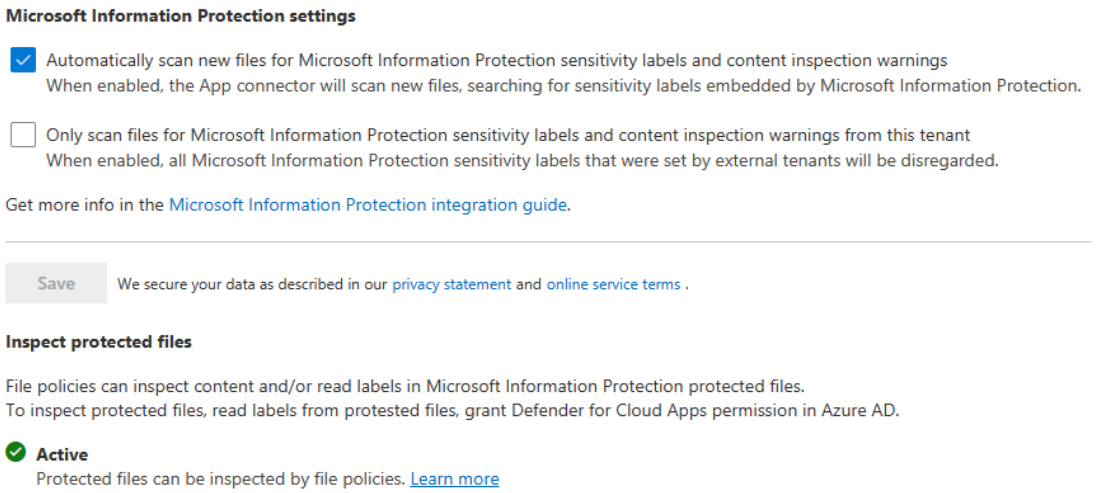
- Enable User Enrichment



Information Protection

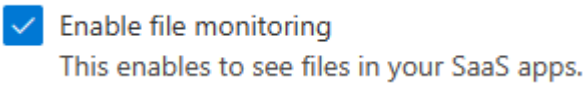
MICROSOFT INFORMATION PROTECTION

- Enable automatically scan new files
- Enable scanning protected files. You'll need to go through the OAUTH grant process.



FILES

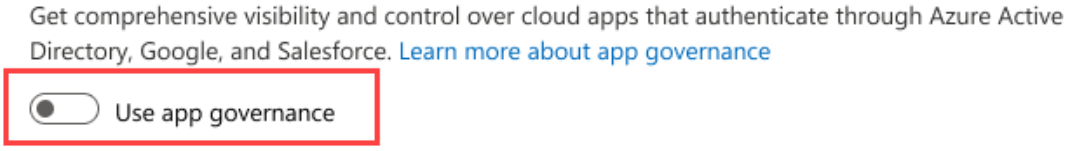
- Enable file monitoring



App governance

SERVICE STATUS

- Turn on app governance



Connected Apps

APP CONNECTORS

- Hit Connect an app, choose Microsoft 365 from the list. Select all options.

Before you connect Office 365, we highly recommend reviewing the Office 365 connection guide. Follow these steps in order to connect Office 365.

To connect this app, provide your access credentials. We secure your data as described in the privacy statement | Terms

- [x] Azure AD Users and groups
- [ ] Azure AD Management events
- [ ] Azure AD Sign-in events
- [ ] Azure AD Apps
- [ ] Azure AD activities
- [ ] Azure AD files

Enable file monitoring before enabling Office 365 files.

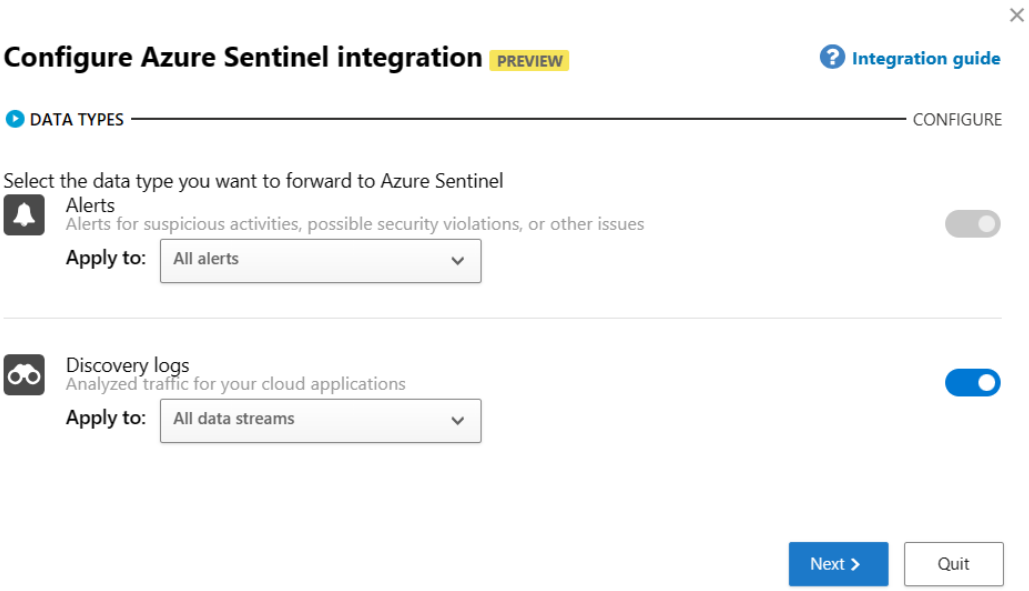
SIEM Agents

- Add the Azure Sentinel integration

SIEM agents

Add SIEM agent

Generic SIEM	Integrate with your SIEM server
Azure Sentinel	Integrate to Microsoft cloud-native SIEM



MDE

In the Defender Portal, go to Settings -> Endpoints

Advanced Features

Set the features configured below

This section provides a set of advanced features you can enable. These features require integration with other products. You need to verify that these settings are enabled to use the features.

Off

Restrict correlation to within scoped device groups

When this setting is turned on, alerts are correlated into separate incidents based on their scoped device group. By default, incident correlation happens across the entire tenant scope.

> ⓘ Changing this setting impacts future alert correlations only.

Off

Enable EDR in block mode

When turned on, Microsoft Defender for Endpoint leverages behavioral blocking and containment capabilities by blocking malicious artifacts or behaviors observed through post-breach endpoint detection and response (EDR) capabilities. This feature does not change how Microsoft Defender for Endpoint performs detection, alert generation, and incident correlation. To get the best protection, make sure to apply security baselines in Intune. See EDR in block mode for more details.

On

Automatically resolve alerts

Resolves an alert if Automated investigation finds no threats or has successfully remediated all malicious artifacts.

Off

Allow or block file

Make sure that Windows Defender Antivirus is turned on and the cloud-based protection feature is enabled in your organization to use the allow or block file feature.

On

Hide potential duplicate device records

When turned on, this setting will hide duplications that might occur for the following reasons:

- Devices that were discovered more than once
- Discovery of onboarded devices
- Unintentionally discovered onboarded devices

These duplications will be hidden from multiple experiences in the portal to create a more accurate view of the device inventory. The affected areas in the portal include the Device Inventory, Microsoft Defender Vulnerability Management screens, and Public API for machines data. Notably, you will still be able to view these devices in global search, advanced hunting and alert and incidents pages.

> ⓘ When activated, this heuristic might hide some discovered devices in certain cases. You can always come back here and choose to view all devices.

On

Custom network indicators

Configures devices to allow or block connections to IP addresses, domains, or URLs in your custom indicator lists. To use this feature, devices must be running Windows 10 version 1709 or later. They should also have network protection in block mode

and version 4.18.1906.3 or later of the antimalware platform (see KB 4052623). Note that network protection leverages reputation services that process requests in locations that might be outside of the location you have selected for your Microsoft Defender for Endpoint data.

On

Tamper protection

Stop unwanted changes to your security solution and its essential functions. With tamper protection, malicious apps are prevented from turning off security features like virus & threat protection, behavior monitoring, cloud-delivered protection, and more. Learn about tamper protection requirements

On

Show user details

Enables displaying user details: picture, name, title, department, stored in Azure Active Directory.

On

Skype for business integration

Enables 1-click communication with users.

Pending

Office 365 Threat Intelligence connection

Connects to Office 365 Threat Intelligence to enable security investigations across Office 365 mailboxes and Windows devices.

For more information, see the Office 365 Threat Intelligence overview.

Cannot turn feature on. This feature requires an Office 365 E5 license or the Threat Intelligence add-on for Office 365.

On

Microsoft Defender for Cloud Apps

Forwards Microsoft Defender for Endpoint signals to Defender for Cloud Apps, giving administrators deeper visibility into both sanctioned cloud apps and shadow IT. It also gives them the ability to block unauthorized applications when the custom network indicators setting is turned on. Forwarded data is stored and processed in the same location as your Cloud App Security data. This feature is available with an E5 license for Enterprise Mobility + Security on devices running Windows 10 version 1709 (OS Build 16299.1085 with KB4493441), Windows 10 version 1803 (OS Build 17134.704 with KB4493464), Windows 10 version 1809 (OS Build 17763.379 with KB4489899) or later Windows 10 versions.

On

Web content filtering

Block access to websites containing unwanted content and track web activity across all domains. To specify the web content categories you want to block, create a web content filtering policy. Ensure you have network protection in block mode when deploying the Microsoft Defender for Endpoint security baseline.

On

Device discovery

Allows onboarded devices to discover unmanaged devices in your network and assess vulnerabilities and risks. For more information, see Device discovery settings to configure discovery settings.

On

Download quarantined files

Backup quarantined files in a secure and compliant location so they can be downloaded directly from quarantine.

On

Live Response

Allows users with appropriate RBAC permissions to investigate devices that they are authorized to access, using a remote shell connection.

Off

Live Response for Servers

Allows users with Live Response privileges to connect remotely to servers (Windows Server or Linux devices) that they are authorized to access.

Off

Live Response unsigned script execution

Enables using unsigned PowerShell scripts in Live Response.

Off

Share endpoint alerts with Microsoft Compliance Center

Forwards endpoint security alerts and their triage status to Microsoft Compliance Center, allowing you to enhance insider risk management policies with alerts and remediate internal risks before they cause harm. Forwarded data is processed and stored in the same location as your Office 365 data.

On

Microsoft Intune connection

Connects to Microsoft Intune to enable sharing of device information and enhanced policy enforcement.

Intune provides additional information about managed devices for secure score. It can use risk information to enforce conditional access and other security policies.

On

Authenticated telemetry

Keep authenticated telemetry turned on to prevent spoofing telemetry into your dashboard

On

Preview features

Allow access to preview features. Turn on to be among the first to try upcoming features.

See the Microsoft Defender for Endpoint preview features section in the Microsoft Defender for Endpoint guide.

Endpoint Attack Notifications

Enables Microsoft to actively hunt for critical threats to be prioritized based on urgency and impact over your endpoint data. For proactive hunting across the full scope of Microsoft 365 Defender including threats that span email, collaboration, identity, cloud applications, as well as endpoints, learn more about Microsoft Defender Experts.

Apply

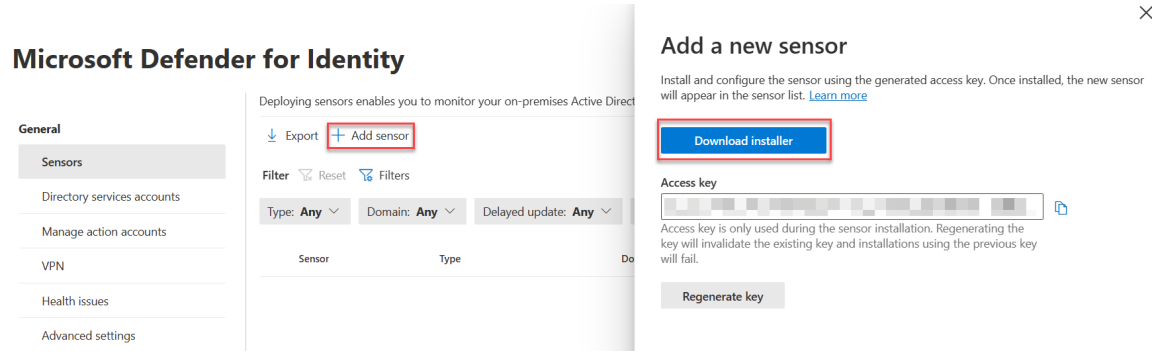
Defender for Identity (MDI)

General

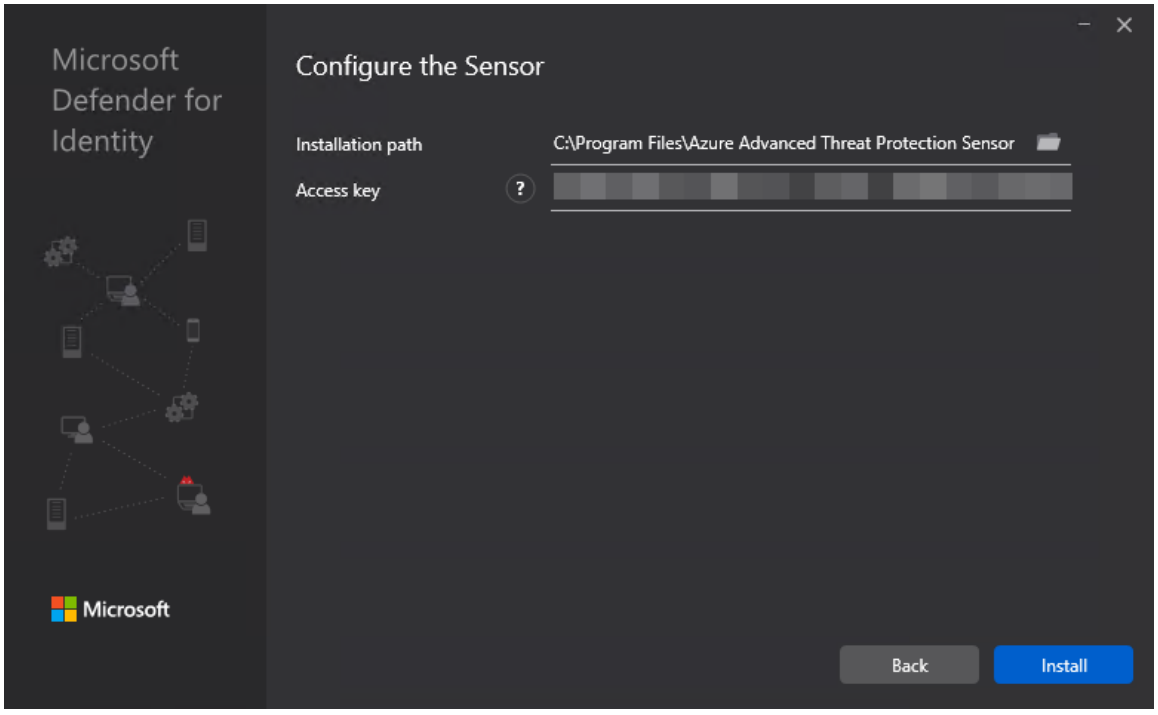
In the Defender Portal, go to Settings -> Identities

SENSORS

Click +Add Sensor, and download the installer and copy the Access key



Install the sensor on all DCs in AD. Use the access key when prompted by the installer.



After installing, configure the Active Directory requirements listed below.

AD Configuration

- Configure Event Collection via GPO
- Configure Group Managed Service Account account
  - On the first DC
    - Create root KDS key `powershell`
      `Add-KdsRootKey -EffectiveTime ((get-date).addhours(-10))`

- Purge kerberos tickets `powershell`
  `klist purge -li 0x3e7`
- Create the gMSA ``` powershell New-ADServiceAccount accountname -PrincipalsAllowedToRetrieveManagedPassword "Domain Controllers" -DNSHostName accountname.domain.contoso.com

```
- Install the gMSA on the DC
  ``` powershell
  Install-ADServiceAccount -Identity 'accountname'
```

- On the other DCs, purge kerberos tickets and install the service account
- Add the gMSA in the portal Adding a gMSA account

Microsoft Purview

Device Onboarding

📖Windows
📖Mac
🔗

1. Enable Windows and Mac device onboarding. This requires MDE.

Sensitivity Labels

Enable labeling for Protected content & PDFs

- Enable Sensitivity Labels for Protected Content
- Enable labeling support for PDFs

With the SharePoint Module in PowerShell 5

```
connect-sposervice -url 'https://<tenant>-admin.sharepoint.com/'
Set-SPOTenant -EnableAIPIntegration $true
Set-SPOTenant -EnableSensitivityLabelforPDF $true
```

Enable Labeling for Containers

In a fresh tenant, there will not be any EntraID group settings configured, so those need to be created. After that, you can enable the label sync.

With the AzureADPreview module in Windows PowerShell (PS5)

```
Connect-AzureAd
$TemplateId = (Get-AzureADDirectorySettingTemplate | where { $_.DisplayName -
```
```

```
eq "Group.Unified" }).Id
$Template = Get-AzureADDirectorySettingTemplate | where -Property Id -Value
$TemplateId -EQ
$setting = $Template.CreateDirectorySetting()
$setting["EnableMIPLabels"] = "True"
New-AzureADDirectorySetting -DirectorySetting $Setting
```

Then, with the ExchangeOnlineManagement module in PS5/PS7+

```
Connect-IPPSSession
Execute-AzureAdLabelSync
```

Enable co-authoring for Encrypted Files

This can be done in the portal, or via PowerShell.

With the ExchangeOnlineManagement module in PS5/PS7+

```
Connect-IPPSSession
Set-PolicyConfig -EnableLabelCoauth:$true
```

DLP

Endpoint DLP

SETTINGS

🔗 Portal 📘 Docs

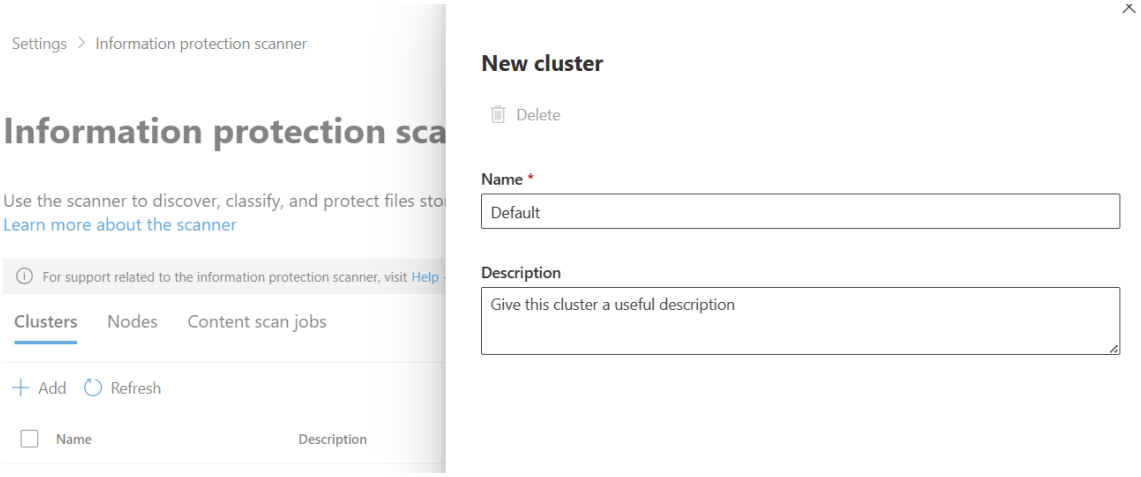
In Settings, change the following: - Advanced classification scanning and protection: `On`

MIP Scanner

🔗 Portal 📘 Docs

Prerequisites - Service account in AD, exclude from MFA registration and CAs - SQL server for the scanner, as well as a windows server.

Deployment 1. Create a Scanner Cluster



2. Create a Content Scan Job. Be sure to disable any of the auto options - this will just be for

scanning.

Edit content scan job

General

Content scan jobs specify how the on-premise repositories should be scanned

Content scan job name *

Default

Description

Cluster

Default

Schedule

Manual

Info types to be discovered

All

Treat recommended labeling as automatic

Off

Enable DLP policy rules

Off

Enforce sensitivity labeling policy

Off

Label files based on content

On

Default label

None

Relabel files

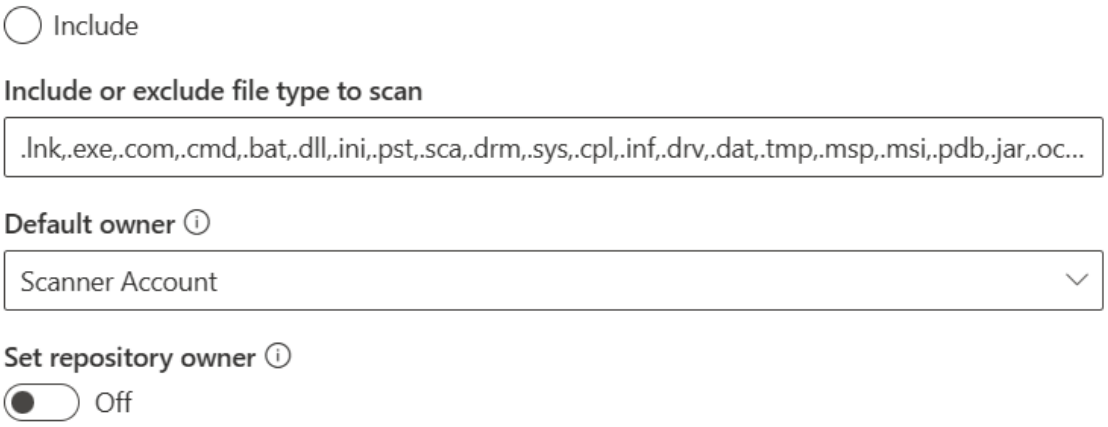
Off

Preserve "Date modified", "Last modified", and "Modified by"

On

Include or exclude file type to scan *

Exclude



Exact Data Match

Portal
Docs

EDM SIT

1. Create a new EDM SIT. Since I work in healthcare, I typically use Synthea to generate patient records. We do have sample industry files you can use.
2. Make note of the datastrore name when you finish the EDM wizard. You'll need it for the EDM uploader.

EDM UPLOADER TOOL

1. Create a service account for the EDM upload agents to run as.
2. Create a EntraID security group named `EDM_DataUploaders` and add the service account to it.
3. Install the EDM upload tool to `c:\EDM`
4. Place sample data in `c:\EDM\Data`
5. Save the schema `.\EdmUploadAgent.exe /SaveSchema /DataStoreName your_data_store_name /OutputDir c:\edm\data`
6. Create `c:\EDM\hash`
7. Upload the data `.\EdmUploadAgent.exe /uploaddata /datastorename your_data_store_name /datafile C:\edm\data\your_data.csv /hashlocation c:\edm\hash /schema C:\edm\data\your_data_store_name.xml /allowedbadlinespercentage 5`

Insider Risk Management

Roles

🔗 Portal
📘 Docs

Add your account to the `Insider Risk Management` role.

Browser Activity Plugins

1. Deploy the Edge profile via Intune as described here
2. Deploy the Chrome profile via Intune as described here

Settings

🔗 Portal
📘 Docs

POLICY INDICATORS

- Select all indicators under the following categories
  - Office
  - Device
  - Microsoft Defender for Endpoint
  - Risky Browsing
  - Microsoft Defender for Cloud Apps

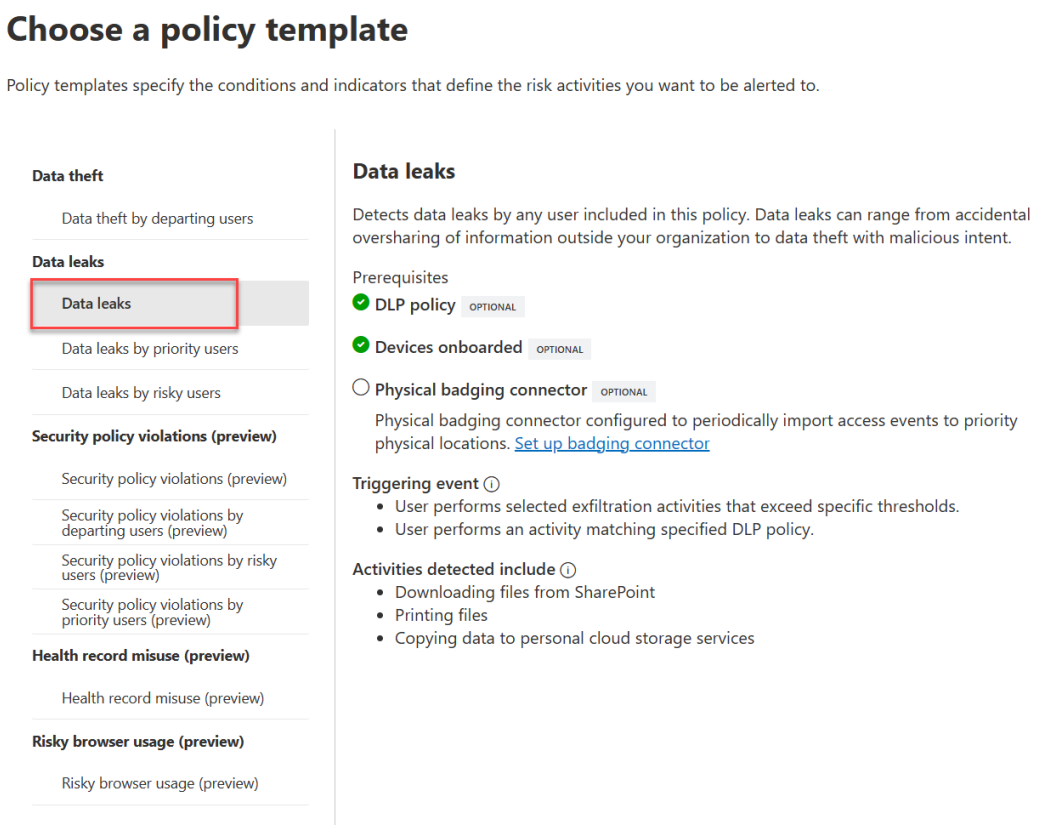
ANALYTICS

- Toggle analytics on

Polices

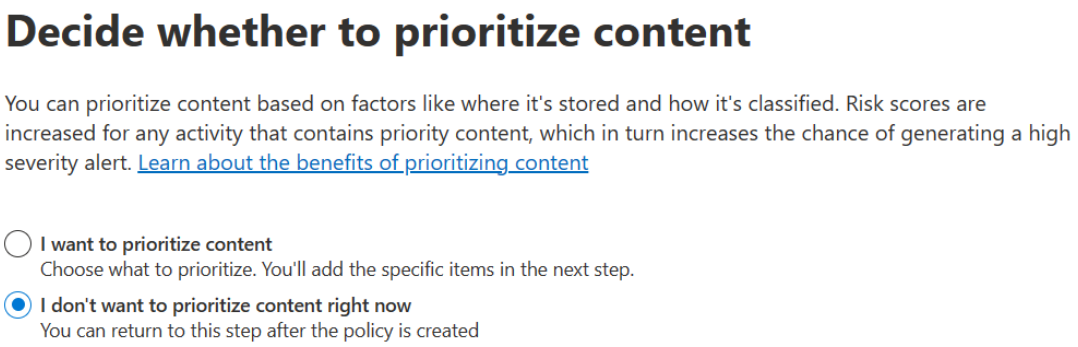
🔗 Portal
📘 Docs

1. Create a Data Leaks policy from the template



2. Target all users

3. Choose not to prioritize content.



4. For triggering events, choose `User performs an exfiltration activity`

5. For thresholds, choose `Apply built-in thresholds.`

6. For indicators, leave the default ones checked.